Stable Frontiers, Uncertain Decisions in a Synthetic Hospital Ransomware Model

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Abstract: The largest change in a simulated outcome need not produce the largest change in a decision. We examine this distinction in a synthetic hospital ransomware model with six objectives and shared scenarios across defense portfolios. Changing the outage threshold moves high-capacity event frequency by 14.34 percentage points without changing its frontier; a smaller frequency change alters the resource-constrained frontier. We then examine the assumptions that this endpoint comparison holds fixed. An interval calculation retains 72 observed efficient portfolios under endpoint uncertainty, falling to 27 when shared component prices vary while upgrades remain strictly costly. Paired resampling and coefficient-conditioned regimes reveal further changes in membership. A comparison of Hoeffding, empirical Bernstein, and approximate paired t bounds shows how the inferential method changes the conclusions supported by the same data. Empirical Bernstein improves the margins but still certifies only free baselines; the approximate method supports additional portfolios under nominal prices. Public clinical-harm records independently expose gaps in modeled services and time coverage. The contribution is an auditable comparison of these sources of decision instability, with a shared-price retention calculation and explicit distinctions between deterministic guarantees, conditional sensitivity, and statistical inference. The findings direct validation toward assumptions that can change the choice, while keeping the limits of the synthetic evidence visible.

Index Terms: **simulation; defense portfolios; Pareto efficiency; shared costs; sampling uncertainty; reproducibility**

1. Introduction

Security investments compete for money, staff time, and operational attention. A simulation can compare many combinations of defenses, but an efficient portfolio is only as persuasive as the assumptions behind its objectives. Reproducing the objective table does not establish that its frontier will survive a different outcome definition, price schedule, or scenario sample.

We investigate these distinctions in a single synthetic hospital ransomware model. The experiment compares 400 profile-candidates on disruption, extreme loss, outage frequency, non-recovery, cost, and burden. An initial check appears reassuring: changing the outage definition leaves two profiles' frontiers unchanged. Yet this check holds both prices and other estimated objectives fixed. Our research question is therefore: *how much of that apparent stability survives changes in endpoint definitions, shared prices, scenario samples, and the assumed effectiveness of controls?*

The answer changes the interpretation of the experiment. The 72 portfolios retained under endpoint uncertainty fall to 27 under a shared-price region that keeps upgrades strictly costly. Scenario resampling provides a further, distinct check; it is not a statistical guarantee. External clinical-harm data reveal where the modeled services and horizon fall short of reported disruption.

The contribution is this joint analysis and its inspectable implementation. Security portfolio simulation-optimization, interval efficiency, and bootstrap selection are established methods [1]–[3]. We combine them in a finite paired experiment, exploit the model's shared component-price structure, and compare the conclusions supported by deterministic, resampling, and population analyses. A further comparison measures how much finite-sample bounds lose relative to approximate paired inference. Coefficient-conditioned frontiers expose changes that separate objective rankings can miss. The hospital reanalysis is retrospective. Its plans record which results were already known and when the positive-upgrade-price extension was added.

2. Model and Experimental Design

2.1. Mechanisms and evidence

The model represents one facility as a network of 200–300 digital assets. Incidents enter through a workstation, internet-facing asset, privileged system, medical device, or vendor connection. Compromise, detection, isolation, and restoration evolve in five-minute steps over 72 hours. The resource-constrained, intermediate, and high-capacity profiles are fixed starting environments. Portfolios add controls to that environment; they cannot buy a different capacity profile.

Controls follow the pathway on which they act. Patching mainly reduces exploit-mediated traversal, has little effect on credential-mediated traversal, and has partial effect on vendor pathways. Identity protection has separate coverage and effectiveness on eligible authentication paths. Isolated backups lack a normal traversal path but can suffer an incident-level isolation lapse or residual recovery failure. Restoration either waits for containment or proceeds in parallel in a separate structural experiment. Restored nodes become susceptible again, allowing reinfection.

Figure 1 separates three levels of the experiment. Network pathways determine which assets can be reached.

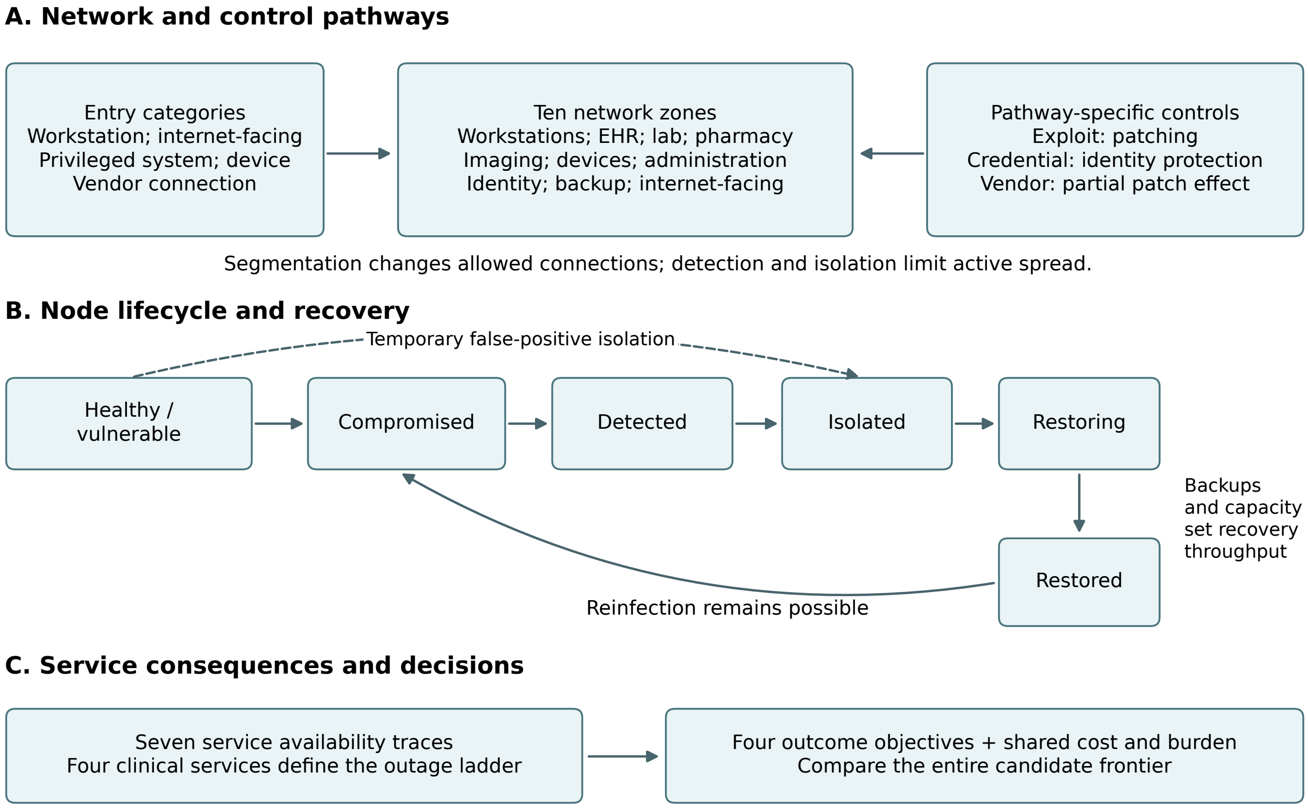

Figure 1. Model structure and control pathways. Panel A groups the network inputs, not a measured hospital topology. Panel B is a conceptual lifecycle: healthy and vulnerable are distinct reporting categories, giving seven categories in total. The engine uses separate compromise, isolation, and restoration flags. False-positive isolation expires; restoration does not confer immunity. Panel C links digital availability to the six objectives.

Detection and response determine how long compromised assets remain active or unavailable. Service aggregation converts asset availability into the quantities used for portfolio selection. A defense can affect these levels differently: isolation can slow spread while also removing an otherwise functioning asset from service. The model therefore evaluates the combined consequences of a portfolio.

Restoration illustrates why mechanism choices matter. Isolated compromised nodes enter a queue ordered by criticality. A throughput allowance determines how many restorations begin, and each takes a fixed number of steps to finish. Unavailable backups reduce that allowance. The primary model gates restoration on containment; the structural alternative starts recovery while active compromise remains elsewhere. Neither rule represents a measured hospital recovery schedule. They are competing mechanisms whose consequences can be compared in a paired experiment.

An evidence register distinguishes observed quantities, contextual evidence, technical assumptions, and normalized decision units. Hospital incident studies establish that disruption occurs [4]; authentication research informs the scope of identity protection [5]. Neither identifies whole-network control effects in this model. Technical effect sizes remain non-identifiable from the available public sources. We sample 10 coefficients over declared ranges once per scenario and share the vector across candidate replays. These ranges express assumptions to examine, not estimates of hospital-sector effectiveness. Appendix D lists every range alongside its frontier sensitivity.

2.2. Objectives and paired scenarios

A service is available when at least 60% of its supporting nodes are functional. Weighted service-hours lost sum unavailable time across EHR, laboratory, pharmacy, imaging, scheduling, identity, and backup/recovery, with weights $(1.0, 0.8, 0.9, 0.7, 0.4, 0.9, 0.6)$. The resulting loss is bounded by 381.6 weighted hours. Only the first four services enter the clinical-outage endpoint. This endpoint measures digital-service availability, not patient outcomes or treatment capacity.

Each candidate has six minimized objectives: mean weighted loss, empirical upper-tail mean loss, sustained-outage frequency, non-recovery by 72 hours, implementation cost, and operational burden. The historical tail statistic averages losses at or above the empirical 90th percentile, including ties. Cost and burden are normalized scenario points. Segmentation, patching, detection, isolation, backups, and identity protection are priced as upgrades above the profile's existing posture. The three profiles have 288, 96, and 16 distinct resolved candidates.

Discovery uses 30 scenarios per profile and 12,000 candidate executions. Confirmation uses 400 fresh scenarios per profile and 160,000 executions. Within a profile, all candidates share event-keyed random inputs. Disjoint scenario-ID blocks separate stages and profiles. Confirmation allocates 80 scenarios to each of five entry categories, defining an equally weighted scenario mixture. The independent units are scenarios; candidate executions are paired observations.

2.3. What a portfolio comparison means

Let A_{si} be the fraction of available nodes supporting service s in time step i, w_s its declared weight, and Δt the step length in hours. The recorded loss is

$$L = \Delta t \sum_i \sum_s w_s \mathbf{1}\{A_{si} < 0.60\}. \quad (1)$$

Thus one weighted service-hour can arise from several patterns of service loss. It is not one patient-hour, one hour of complete hospital closure, or a monetary loss. The 72-hour endpoint also censors longer disruption. These definitions make the arithmetic reproducible while leaving the clinical interpretation dependent on evidence outside the simulator.

The six-objective frontier answers a restricted question: which enumerated portfolios have no observed competitor that is at least as good on every objective and better on one? It does not select one preferred portfolio. Cost and burden remain separate because inexpensive controls can consume staff effort, while a low-burden choice can require greater expenditure. Turning these tradeoffs into a selection would require an explicit budget, burden limit, or preference rule. The budget example in Appendix B illustrates one such rule without treating it as a universal recommendation.

The full candidate space is small enough to enumerate. This avoids conflating failure to find a candidate with evidence that it is inefficient. It also lets us compare the frozen discovery shortlist with the full confirmation frontier on the same scenario bank. The three capacity profiles define separate feasible sets and baselines; a resource-constrained candidate is never removed because a candidate from another profile dominates it.

3. Stability Analysis

3.1. Endpoint definitions and interval efficiency

Let N_{pci} count clinical services whose individual continuous outage exceeds two hours for profile p, candidate c, and scenario i. Define $Y_{pci}^{(k)} = \mathbf{1}\{N_{pci} \geq k\}$ for $k = 1, \ldots, 4$. The qualifying outages need not overlap. We report all four values of k. The primary endpoint uses $k = 4$. For a fixed population and weighting,

$$\overline{Y^{(1)}} - \overline{Y^{(4)}} = \overline{\mathbf{1}\{1 \leq N \leq 3\}}. \quad (2)$$

The partial-outage fraction is exactly the threshold sensitivity. Using it to justify threshold insensitivity would be circular.

Write $a \prec b$ if a is no worse in every minimized coordinate and strictly better in at least one. The efficient set F contains candidates with no dominator. For independent bounds $L_c \leq f_c \leq U_c$, standard necessary/possible efficiency gives the finite sets

$$G = \{c : \nexists d \neq c,\ L_d \prec U_c\}, \quad (3)$$
$$P = \{c : \nexists d \neq c,\ U_d \prec L_c\}. \quad (4)$$

Thus $G = \bigcap F$ and $P = \bigcup F$ over the full Cartesian box [2]. Appendix A gives the short proof. The initial application varies each candidate's outage frequency from its $k = 4$ to $k = 1$ estimate while holding five objectives fixed. This box includes every common-k specification and additional candidate-specific combinations.

3.2. Shared cost and burden uncertainty

Independent candidate cost intervals would discard the model's pricing structure. Instead, write cost as $a_c^\top t$ and burden as $a_c^\top u$, where a_c records purchased upgrades and t, u are shared eight-component price vectors. Upgrade coefficients can be negative when subtracting the price of a baseline ladder rung. Every candidate and profile uses the same component prices within each family.

For radius ρ, prices lie between $(1 - \rho)$ and $(1 + \rho)$ times their nominal values. Basic segmentation cannot exceed least-privilege segmentation, and periodic backups cannot exceed protected backups. We call this the *monotone region*. A second region also requires each of these ladder gaps to retain at least $(1 - \rho)$ times its nominal incremental price. This *positive-gap region* excludes free upgrades. Both are declared stress regions. We evaluate $\rho \in \{0, 0.10, 0.25, 0.50, 0.75\}$.

For each rival d and candidate c, compute

$$m_{dc}^t = \min_{t \in T_\rho} (a_d - a_c)^\top t, \qquad m_{dc}^u = \min_{u \in U_\rho} (a_d - a_c)^\top u. \quad (5)$$

Each price region separates into four scalar intervals and two constrained rectangles. Evaluating their vertices gives the exact linear minimum. Combine these minima with the lower differences in the four stochastic objectives. A rival can dominate under some admissible realization exactly when all six minima are nonpositive and at least one is negative. A candidate is guaranteed efficient when no rival satisfies that condition.

This calculation preserves shared prices and takes $O(n^2 m)$ arithmetic operations for n candidates, m objectives, and fixed price dimension. For each rival, its outcome, cost, and burden minima can be attained together because those three uncertainty blocks are independent. That suffices for guaranteed retention. It does not produce an exact possibly efficient set under coupled prices: avoiding each rival at a different price need not avoid all rivals at one price.

3.3. Resampling and simultaneous mean bounds

The deterministic calculation fixes estimated outcomes. We next resample whole scenarios within each entry category, preserving candidate pairing. Each of 1000 draws per profile recomputes all four stochastic objectives and both endpoint extremes. We apply every price radius and region to those same draws. Candidate retention frequencies and count quantiles describe this conditional resampling experiment. We do not interpret the discontinuous frontier's bootstrap percentiles as confidence sets for population efficiency.

For a separate finite-sample check, define paired rival-minus-candidate differences in mean loss, non-recovery, and the conservative outage contrast $Y_d^{(4)} - Y_c^{(1)}$. Their ranges have widths $2M, 2, 2$, where $M = 381.6$. Hoeffding's inequality and a union bound [6] give simultaneous one-sided lower bounds

$$\ell_{dc,j} = \overline{D}_{dc,j} - R_j \sqrt{\frac{\log(K/\alpha)}{2n}}, \tag{6}$$

where R_j is the range width, $n = 400$, and $K = 276,048$ includes three means for every ordered distinct pair across all profiles. Independent scenarios with the fixed stratum allocation suffice; comparisons need not be independent. We report $\alpha = 0.10, 0.05, 0.01$, each defining its own simultaneous family.

We certify a candidate only if every rival has a strictly positive lower bound in at least one of these means or in Eq. 5. The rival then cannot dominate, whatever the tail coordinate is. Requiring a strictly worse rival coordinate is essential: ties in all screened coordinates leave room for dominance on the omitted tail objective. This is a sufficient, potentially conservative population-efficiency screen for the declared scenario generator.

3.4. Variance-sensitive comparison

The range-only margin in Eq. 6 cannot exploit a small variance of paired differences. We compare it with the empirical Bernstein bound of Maurer and Pontil [7]. Their Theorem 11 allows independent observations with different distributions, matching the fixed entry allocation here. With pooled sample variance s_D^2, its margin is

$$b_{\mathrm{EB}} = \sqrt{\frac{2s_D^2 \log(2K/\alpha)}{n}} + \frac{7R\log(2K/\alpha)}{3(n-1)}. \tag{7}$$

The lower bound is $\overline{D} - b_{\mathrm{EB}}$. Pairing enters through s_D^2, including the covariance between candidates. The second term remains even when the observed differences are constant. The pooled variance may also include differences between entry-category means; the theorem permits this conservatism without treating the allocation as an i.i.d. mixture sample.

For an approximate comparison, we use the within-entry paired variance in Eq. 8, Welch–Satterthwaite degrees of freedom ν, and margin $t_{\nu,1-\alpha/K}\widehat{\mathrm{SE}}(\overline{D})$. An estimated variance of zero triggers the Hoeffding margin, avoiding an unsupported zero-width interval. This is a numerical safeguard, not a finite-sample coverage theorem for the approximate method.

All three methods use the same $K = 276,048$ family at $\alpha = 0.05$, the same paired observations, and the same sufficient population screen. We report nominal prices and the positive-gap region at $\rho = 0.50$. We also count positive lower bounds on mean-loss improvement against the free baseline. These answer different questions: a positive mean-loss difference need not establish six-objective efficiency. Each method receives its own error allocation. Selecting the smallest realized margin across methods would require an additional multiplicity argument.

3.5. Coefficient-conditioned frontier analysis

The original coefficient analysis compares separate objective rankings across low and high parameter thirds. We extend that question to the complete decision set. For each of the ten recorded coefficients, profile, and entry category, we select the lowest and highest 26 of its 80 scenario draws. Each regime contains 130 paired scenarios with equal entry-category weights. Stable scenario order resolves ties. Every candidate is evaluated through its recorded outcomes on that same subset; no simulator coefficient is changed.

For each of the resulting 60 regimes, we recompute the six objectives, including the empirical tail with its original ties convention. We measure frontier Jaccard similarity to the full-bank frontier, additions, removals, and endpoint and shared-price certificates. Jaccard is the size of the intersection divided by the size of the union. A low value identifies a different decision set, even if individual objective rankings remain correlated.

Reducing a bank from 400 to 130 scenarios can itself change its frontier. We therefore generate 1,000 matched-size random subsets per profile, drawing 26 scenarios without replacement within each entry category. Their Jaccard quantiles provide a descriptive reference for the coefficient regimes. The reference uses the same bank, candidate space, and objective estimators. It is not a null distribution for a significance test, and its central range is not a confidence interval. Coefficients other than the selected one, and the realized incident trajectories, can differ between regimes. This analysis measures conditional sensitivity within the recorded design, not the causal effect of setting a coefficient or a guarantee over their joint range.

3.6. Structural and external checks

A distinct paired structural bank of 150 scenarios per profile compares containment-gated and parallel restoration for 96 selected candidates. These alternatives examine mechanisms that resampling a fixed bank cannot change.

We also audit construct coverage against CIPHER v1.0.1, an independently authored public dataset of reported clinical harms [8]. Before aggregation, we mapped its technical-domain labels to represented, absent, or unspecified model services. We count coded records and distinct reference

TABLE 1. CONFIRMATION EVENT FREQUENCIES (%), THEIR GAP IN PERCENTAGE POINTS, AND FRONTIER JACCARD SIMILARITY BETWEEN $k = 1$ AND $k = 4$.

Profile	$k = 1$	$k = 4$	Gap	Jaccard
Resource-constrained	87.32	83.45	3.87	0.896
Intermediate	57.39	44.79	12.60	1.000
High-capacity	16.00	1.66	14.34	1.000

TABLE 2. ENDPOINT INTERVAL CALCULATION WITH FIVE OBJECTIVES FIXED. G: GUARANTEED EFFICIENT; F_4: OBSERVED FRONTIER; P: POSSIBLY EFFICIENT.

Profile	Candidates	$\lvert G\rvert$	$\lvert F_4\rvert$	$\lvert P\rvert$
Resource-constrained	288	42	64	79
Intermediate	96	21	22	41
High-capacity	16	9	9	16

links, correct one documented time-label typo, and report all categories and leave-one-reference-out influence ranges. Week 2 and First Month annotations extend beyond the model horizon; First Week straddles it. These annotations are not recovery durations. CIPHER records are not independent incidents, and its impact scores are not used as model weights.

Two published engineering problems, RE21 and RE22 [9], provide separate implementation checks for Eqs. 3 and 4. Their pre-recorded design uses 128 finite sampled candidates per problem, three objective-bound radii, an independent frontier oracle, and exact small corner enumerations. The shared-price support function is separately checked against linear programming, including positive-gap regions. Appendix B reports the benchmark settings.

4. Results

4.1. Endpoint agreement

The intermediate and high-capacity profiles have double-digit changes in event frequency but identical frontier membership across all four thresholds. The resource-constrained frontier changes despite its smaller frequency gap (Table 1). The size of an outcome change therefore does not rank the size of a decision-set change.

Across the full endpoint box, 72 of 95 observed efficient candidates remain efficient (Table 2). All 9 high-capacity members are retained, while 7 other candidates could join them. Even here, retention is a weaker conclusion than exact frontier invariance.

4.2. Joint price and sampling sensitivity

Adding shared price uncertainty reduces guaranteed retention in every profile (Figure 2). At $\rho = 0.50$, the monotone region retains 21 candidates. Requiring positive upgrade gaps retains 27 (Table 3). The decline is therefore not explained solely by free upgrades at a price-region boundary. The intermediate profile is particularly sensitive to which incremental prices the region permits.

TABLE 3. RETENTION AT $\pm 50\%$ SHARED PRICES. G USES THE ORIGINAL SCENARIO BANK; B_{95} COUNTS ORIGINAL ENDPOINT CERTIFICATES RETAINED IN AT LEAST 95% OF RESAMPLES.

	Monotone		Positive gaps	
Profile	G	B_{95}	G	B_{95}
Resource-constrained	15	5	16	5
Intermediate	1	1	6	5
High-capacity	5	2	5	2

When the stochastic objectives are recomputed from resampled scenarios, 8 original certificates meet the 95% retention-frequency criterion in the monotone region and 12 in the positive-gap region. All radii, per-candidate frequencies, and count distributions are released, including candidates that gain efficiency in some resamples. This result describes which original certificates recur, not a new shortlist of clinically validated defenses.

The simultaneous mean screen is much more conservative. At family $\alpha = 0.05$, its margins are 106.3 weighted hours for loss differences and 27.9 percentage points for binary differences. At $\rho = 0.50$ it certifies 2 portfolios in the monotone region and 3 in the positive-gap region. Those retained are free baselines, protected by their cost and burden coordinates; no purchased portfolio is certified. This remains so at the other two reported alpha levels. The available bank supports useful comparisons, but this distribution-free screen does not resolve their population frontier membership.

The earlier 500 random-price draws also change frontier membership in every profile. The new calculation answers the stronger universal question over the specified price region rather than relying on sampled price draws.

4.3. What variance-sensitive bounds resolve

Empirical Bernstein narrows the high-capacity best-versus-baseline margin from 106.32 to 74.31 weighted hours. The remaining margin is still much larger than the 1.95-hour observed benefit. Its finite-sample range term prevents the small paired variance from producing a narrow interval. Both finite-sample methods certify only the free baselines under the tested price regions.

The approximate paired t margin for that same contrast and comparison family is 1.74 hours, giving a positive lower benefit of 0.21 hours. Across the profiles, its population screen retains 12 purchased candidates at nominal prices, falling to 1 under positive-gap price stress (Table 4). This comparison separates two issues that the original range-only bound could not: the conservatism of the inferential procedure and the sensitivity of relative costs. The additional retentions depend on approximate inference and are not promoted to exact certificates.

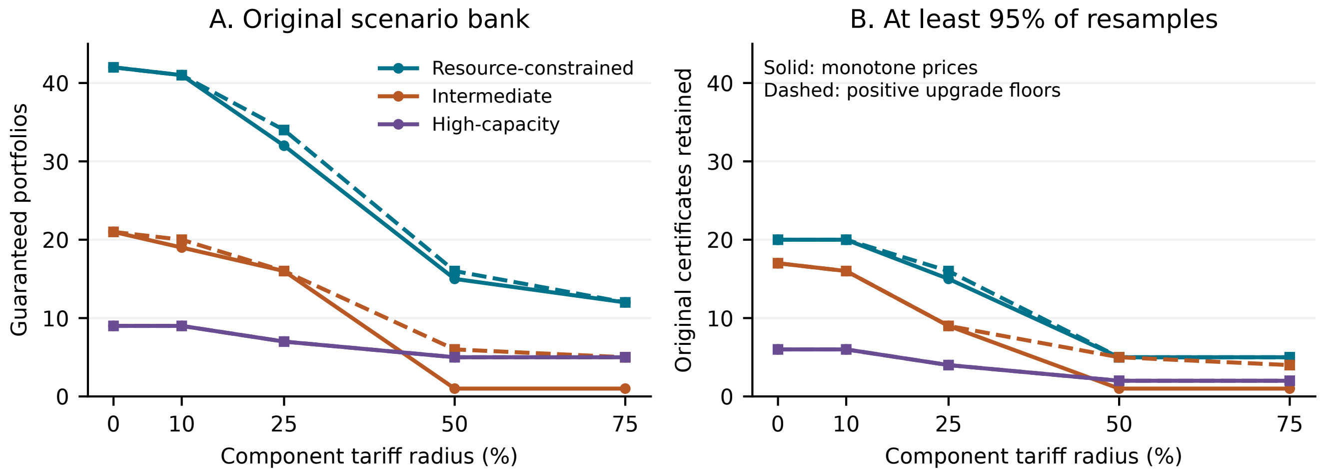

Figure 2. Joint endpoint and price sensitivity across all planned radii. Panel A holds the original outcome estimates fixed. Panel B counts members of the original endpoint-guaranteed set retained in at least 95% of 1000 paired stratified resamples. The percentage in panel B is a resampling frequency, not a confidence level. Positive-gap prices preserve at least $(1 - \rho)$ of each nominal ladder increment.

TABLE 4. Same-family statistical comparison at $\alpha = 0.05$ and $K = 276,048$. The margin is for the largest observed mean-loss benefit against the free baseline, in weighted hours. Positive comparisons count purchased candidates whose mean-loss lower benefit is positive. The final columns count purchased candidates retained by the sufficient population screen at nominal prices and at positive-gap $\rho = 0.50$ prices. The paired t row is approximate; the other rows have finite-sample coverage.

Profile	Method	Margin	Positive comparisons	Retained, $\rho = 0$	Retained, $\rho = .5$
Resource-constrained	Hoeffding	106.32	150	0	0
Resource-constrained	Empirical Bernstein	99.46	150	0	0
Resource-constrained	Paired t (approx.)	24.53	240	5	1
Intermediate	Hoeffding	106.32	53	0	0
Intermediate	Empirical Bernstein	100.57	58	0	0
Intermediate	Paired t (approx.)	25.56	92	6	0
High-capacity	Hoeffding	106.32	0	0	0
High-capacity	Empirical Bernstein	74.31	0	0	0
High-capacity	Paired t (approx.)	1.74	8	1	0

TABLE 5. Coefficient-regime Jaccard ranges and matched-size reference ranges. The last column counts regimes below the reference's 2.5th percentile. It is descriptive and has no multiple-testing interpretation.

Profile	Regimes	Reference	Below
Resource-constrained	0.535–0.815	0.618–0.829	2
Intermediate	0.708–1.000	0.778–1.000	3
High-capacity	0.429–1.000	0.444–1.000	1

4.4. Control assumptions and the complete frontier

The coefficient-conditioned frontiers vary substantially (Figure 3). Their Jaccard similarities range from 0.535 to 0.815 in the resource-constrained profile, 0.708 to 1.000 in the intermediate profile, and 0.429 to 1.000 in the high-capacity profile. The high-capacity range is wide even though its common-threshold frontier was unchanged. Threshold agreement alone therefore misses sensitivity visible elsewhere in the same scenario design.

In the resource-constrained high patch-effect regime, 26 of 64 original frontier members disappear. High traversal rates also give a Jaccard below every matched-size random draw in that profile. Across all profiles, 6 of 60 regimes fall below their reference's 2.5th percentile. We report the complete set in Appendix D, including regimes well inside the reference range.

These comparisons strengthen the sensitivity analysis without identifying any hospital coefficient. In particular, a high-capacity regime's small frontier can change markedly after only a few membership changes, and random subsets also give a broad similarity range. Conditional selection reduces sample size and changes the realized mix of other inputs. The matched-size reference helps expose that limitation; it does not remove it or turn the observed associations into control-effect estimates.

4.5. Structural and clinical gaps

The structural experiment finds 21 candidates efficient only with containment-gated restoration and 3 only with

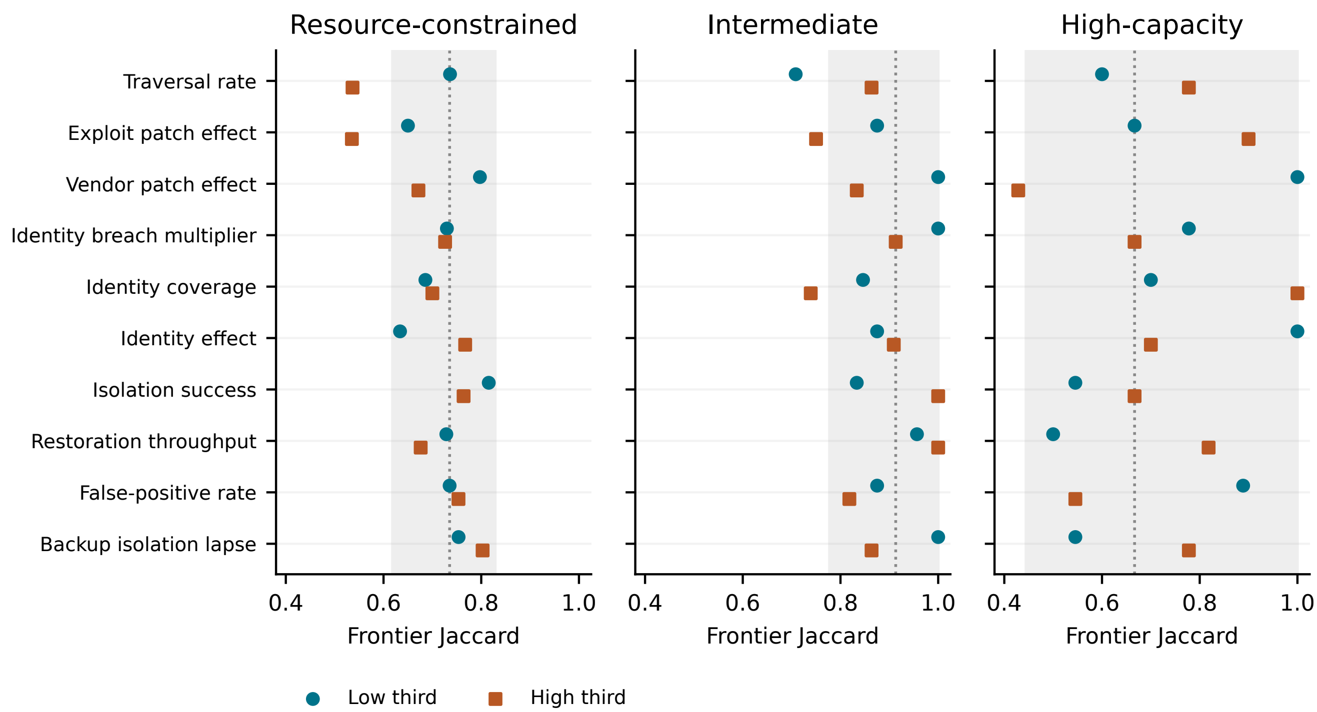

Figure 3. Frontier similarity in coefficient-conditioned regimes. Each point uses 130 paired scenarios, selected within entry categories. The shaded band is the central 95% of Jaccards from matched-size random subsets of the same bank; the dotted line is their median. These bands describe finite-bank variability, not confidence intervals. The two points per row are low and high empirical thirds.

TABLE 6. CIPHER CONSTRUCT COVERAGE. COUNTS DESCRIBE CODED RECORDS, NOT INDEPENDENT INCIDENTS. A REFERENCE CAN CONTRIBUTE TO SEVERAL ROWS.

Coverage class	Records	References	%
Represented service	229	32	72.5
Absent explicit service	85	19	26.9
Unspecified	2	1	0.6

parallel restoration, within the selected candidate bank. Intermediate mean loss rises from 46.93 to 107.41 weighted hours when restoration can begin before containment. Earlier restoration can expose susceptible nodes to a still-active compromise process. High-capacity frontier membership stays unchanged, although objective values move. Parameter ranking correlations, whose worst evaluated value is 0.606, do not settle this structural question.

CIPHER contains 316 records linked to 36 distinct references. Of these, 85 (26.9%) concern technical domains without an explicit modeled service, including communications, infrastructure, operating rooms, and telemetry (Table 6). Another dimension is time: 97 records (30.7%) carry Week 2 or First Month annotations. Removing one reference at a time gives shares of 25.7–30.3% for absent services and 17.4–31.7% for these later annotations. The latter range shows substantial source concentration. Even a represented service label does not establish that its associated patient harm is modeled.

The existing external comparison register also exposes a timing mismatch. The restoration model is slower than the short technical interruption in the CrowdStrike comparison [10] and cannot reproduce the extended clinical disruption in ransomware recovery records [11]. These targets measure different processes, so they cannot be pooled into one fitted restoration rate. Of nine registered comparisons, one is a numeric failure, two are qualitative failures, one shows loose numeric consistency, three are not scored, and two are structurally unaddressable. None establishes calibrated hospital recovery dynamics.

4.6. Interpretation checks

Paired analysis and candidate-set checks resolve two additional errors. The high-capacity candidate with lowest observed mean loss saves 1.95 weighted service-hours against its baseline. Its paired stratified SE is 0.333, versus 0.944 with covariance omitted. The approximate simultaneous interval across all 397 baseline contrasts is [0.66, 3.25] hours. This supports a within-model mean difference without certifying the candidate's six-objective population efficiency.

The full-space frontier has 95 candidates, compared with 57 within the frozen discovery shortlist, which omits 38 efficient candidates. Every resample contains an omission; the rounded central range is 29–49. The original bank has no candidate efficient only because of shortlist restriction, but at least one such artifact appears in 42.7% of resamples.

Finally, the preserved numeric audit passes on three historical interpretation errors. Explicit assertions reject all 3 cases, accept 8 correct controls, and reject or flag invalid all 8 recorded mutations. Table 7 provides the worked examples; Appendix C describes the checks. This is a regression evaluation of specified assertions, not a measure of unseen-error detection.

5. Relation to Prior Work

Security portfolio simulation-optimization already combines control models, multiple objectives, and resource constraints. Kiesling et al. provide a direct precedent [1]. Our contribution concerns the stability of the conclusions drawn from one enumerated, paired experiment. It is not the introduction of simulation-based security portfolio selection.

Efficiency under uncertain coefficients also has a long history. Bitran studies linear multiple-objective problems with interval coefficients [12]; Ide and Schöbel review the distinct notions of robustness used in uncertain multi-objective optimization [13]. Hladík examines possible efficiency and interval robust counterparts [2]. Our finite-box formulas specialize this lineage to a recorded objective table. The shared-price calculation retains the dependence created by a common tariff schedule. Its exactness concerns the stated finite table and uncertainty region, not a new general robustness concept. The price region is a box with ordering and gap constraints; it does not impose a budget on the number of components that can depart from nominal.

Simulation ranking and selection provides a second comparison. Currie and Monks's BootComp uses bootstrap comparisons, permits common random numbers, and includes a hospital ward application [3]. Andradóttir and Lee develop Pareto-set selection procedures with guaranteed correct-selection probabilities under independent or correlated sampling [14]. Our fixed-bank analysis does not replace those selection procedures. It asks how endpoint and shared-price uncertainty interact with uncertainty in already collected simulation outputs. The Hoeffding and empirical Bernstein screens apply established inequalities [6], [7]; the paired t comparison exposes the practical cost of demanding a finite-sample guarantee in this experiment.

The reporting follows STRESS guidance [15] and separates recorded plans from retrospective extensions [16]. Multiverse and specification-curve studies motivate examining analytical choices [17], [18], though we do not implement their joint inference procedures. Statistical consistency checks [19] and metamorphic simulation tests [20] motivate explicit verification of relationships among results. Recent preprints on common random numbers [21] and experimental fidelity [22] provide additional context. The paired design and interpretation checks do not depend on either preprint's priority or empirical claims.

The transferable result is an analysis pattern: distinguish changes in outcome levels from changes in the choice set; preserve shared inputs when varying prices and scenarios; and report what each uncertainty procedure actually resolves. Applying that pattern to this model turns an apparently stable endpoint comparison into a more demanding test of portfolio stability.

6. Discussion

The central result is the mismatch between the size of an outcome change and the size of a decision-set change. The high-capacity endpoint frequency moves substantially while its frontier remains fixed. The resource-constrained frontier changes under a smaller frequency shift. Sensitivity should therefore be evaluated in the quantity that will support the decision, as well as in the numerical outcomes reported beside it.

The shared-price and coefficient analyses identify different reasons for instability. Prices can change a portfolio's relative attractiveness even when its simulated incident outcomes stay fixed. Coefficient-conditioned regimes change the outcome table itself. In the resource-constrained profile, high patch effectiveness and high traversal rates each produce frontier changes beyond the matched-size reference range. These are useful priorities for mechanism validation, not estimates of what stronger patching would cause in a hospital. A targeted intervention experiment would fix those coefficients and vary independent incident scenarios; the present conditional subsets do not make that separation.

The statistical comparison also sharpens the interpretation of a failed certificate. Empirical Bernstein is variance-sensitive, yet its range term remains large under the declared bound and all-pair comparison family. The approximate paired method resolves more nominal-price comparisons, but most of its additional population-screen retentions disappear under price stress. Both sampling uncertainty and valuation assumptions matter. Failure of the finite-sample screen establishes a limit of that procedure with this bank; it does not establish that the underlying population frontier is unknowable.

External evidence exposes a different limitation. Short technical restoration, prolonged clinical disruption, and a time-annotated harm record measure different processes. The available sources do not identify the transition rates that generate those processes here. CIPHER supplies an independently authored construct check, not an independent defense experiment. The single-facility graph also excludes correlated multi-hospital incidents and regional spillover. Applied validation would require measured service dependencies, containment and restoration timelines, and incremental implementation costs.

Several boundaries follow directly from the design. Endpoint intervals allow candidate-specific combinations beyond a common threshold. Price ranges are analyst-chosen stress assumptions. Resampling and coefficient regimes use the same observed bank, and the restoration experiment covers selected candidates. The finite-sample bounds assume independent scenarios; the approximate t screen adds distributional approximation. Table 13 collects these distinctions

TABLE 7. WORKED INTERPRETATION AUDIT. HISTORICAL SENTENCES ARE SHORTENED FAITHFULLY FROM THE ARCHIVED MANUSCRIPT. ASSERTIONS ENCODE THE STATED SCOPE, UNIT, AND BOUND; A PASSING NUMERIC-MACRO CHECK ALONE DID NOT TEST THESE RELATIONS.

Historical statement	Numerical or scope evidence	Executable assertion	Corrected conclusion
Endpoint values move by roughly three percentage points across k.	High-capacity confirmation gap: 14.34 percentage points.	All confirmation profiles present; $\max_p \Delta_p \leq 4$ pp. Fails.	Frequency changes are profile-specific; unchanged frontiers do not imply small frequency gaps.
The high-capacity panel spans under one service-hour.	Baseline-to-best mean difference: 1.95 weighted service-hours.	High-capacity mean-loss range < 1 hour. Fails.	The observed baseline-to-best span exceeds one hour; paired uncertainty is reported separately.
The figure presents discovery results.	Plotted values come from the confirmation objective table.	Source stage equals discovery. Fails.	Identify confirmation as the plotted stage and report discovery separately.

so they can be checked without reading every result as a separate qualification. The free baselines remain an instructive example: a portfolio can be efficient because it costs nothing, without providing an acceptable level of protection.

The work uses synthetic trials and public sources, with no recruited participants or private patient records. The CIPHER archive retains its data license. The earlier THREAT analysis uses a normalized repository preview export, not the original deposited file, and remains descriptive scope evidence. Published engineering functions test implementation on independently defined problems; the evaluation itself remains within this research workflow. Independent replication and transfer to operational hospitals remain open scientific questions beyond the completed artifact.

7. Conclusion

A stable endpoint frontier can coexist with an unstable portfolio decision. In this synthetic hospital model, the largest threshold-driven frequency change leaves the frontier unchanged, while a smaller change alters another profile's frontier. Shared prices reduce endpoint-only retention from 72 to 27 portfolios even when upgrades remain costly. Coefficient-conditioned regimes reveal additional changes in the complete decision set. Comparing finite-sample bounds with approximate paired inference makes the statistical resolution of these claims explicit. The resulting artifact offers a reproducible way to locate the assumptions that constrain a decision, with clinical construct gaps and unresolved mechanism validation reported alongside the computational results.

Artifact Availability

The artifact contains the original trial banks and frozen protocols, retrospective plans and amendments, licensed external inputs, analysis code, generated result tables and figures, and verification commands. The confirmation bank is stored compressed and restored losslessly. Source, data, and manuscript: https://github.com/Dodgeboi/NazeResearchNSRI.

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LLM Usage Statement

AI tools assisted with literature review, code development and revision, verification, and manuscript preparation.

Appendix A. Efficiency Proofs and Numerical Scope

For Eqs. 3 and 4, if $L_d \prec U_c$, placing d at its lower bound and c at its upper bound makes c dominated. Conversely, a realized dominator implies $L_d \prec U_c$. If $U_d \prec L_c$, d dominates c throughout the box. Otherwise, place c at L_c and every rival at its upper bound; no rival dominates c. Independent box coordinates permit these assignments. Strict dominance retains exact ties.

For Eq. 5, minimizing a linear function on a constrained rectangle requires only its vertices: admissible box corners and the intersections with the minimum-gap boundary. Uncoupled coordinates choose the lower or upper endpoint according to the coefficient's sign. Cost, burden, and outcome minima can be realized simultaneously for a fixed rival. The existence of any such rival disproves universal retention; its absence proves retention. Price calculations use integer upgrade coefficients and round cancellation residues to 12 decimal places. Independent linear-programming checks agree to 10^{-10}. The decimal-price implementation constructs intersections in the original coordinates and admits boundary roundoff within eight machine epsilons at the input scale. This fixes feasible singleton regions such as adjacent prices 0.1 and 0.4. Regression cases cover small and large scales and reject gaps that are truly infeasible. The regenerated integer-tariff analysis reproduces every published joint-analysis CSV byte for byte.

TABLE 8. All price radii. Entries give counts in R/I/H profile order. B_{95} counts original certificates meeting the resampling-frequency criterion. The positive-gap extension was recorded after the first run.

	Monotone		Positive gaps	
Radius %	G	B_{95}	G	B_{95}
0	42/21/9	20/17/6	42/21/9	20/17/6
10	41/19/9	20/16/6	41/20/9	20/16/6
25	32/16/7	15/9/4	34/16/7	16/9/4
50	15/1/5	5/1/2	16/6/5	5/5/2
75	12/1/5	5/1/2	12/5/5	5/4/2

TABLE 9. All planned engineering settings, with 128 finite sampled candidates per problem. Radius scales the sampled objective range.

Problem	Radius	$\lvert G\rvert$	$\lvert F_0\rvert$	$\lvert P\rvert$
RE21	1%	9	15	19
RE21	5%	0	15	65
RE21	10%	0	15	120
RE22	1%	0	12	91
RE22	5%	0	12	120
RE22	10%	0	12	125

The population screen follows from Hoeffding's inequality for independent bounded observations, including non-identically distributed observations in fixed strata. A union bound over K comparisons gives simultaneous coverage of at least $1 - \alpha$. If a lower bound is strictly positive for every rival in some coordinate, none can dominate the candidate. No tail-mean concentration claim is required. An equal screened vector cannot be discarded, since an omitted tail coordinate could break the tie.

Appendix B. Engineering and Decision Diagnostics

RE21 models a four-bar truss; RE22 models a reinforced-concrete beam and retains its constraint-violation objective [9]. Independent intervals perturb all objectives by $\pm 1\%, \pm 5\%, \pm 10\%$ of their finite sampled ranges, with non-negative lower bounds. All 6,000 perturbation frontiers contain G and lie within P. For the first eight candidates per problem, 512 total corner matrices with only the second objective uncertain reproduce the exact frontier intersection and union. 5 of six settings have empty guaranteed sets. These checks concern finite tables and chosen bounds, not the continuous problems' global frontiers.

TABLE 10. ILLUSTRATIVE MEAN-LOSS SELECTION WITHIN EACH DECLARED POINT BUDGET. FREE AND SELECTED COLUMNS GIVE MEAN WEIGHTED SERVICE-HOURS LOST; THE LAST COLUMN GIVES THE SELECTED CANDIDATE'S $k = 4$ FREQUENCY.

Profile	Budget	Free	Selected	Event %
Resource-constrained	5	322.13	156.99	94.50
Intermediate	10	179.34	24.34	21.50
High-capacity	15	15.02	13.07	0.00

TABLE 11. FRONTIER SIZES OVER EVERY OBJECTIVE SUBSET OF A GIVEN DIMENSION, SHOWN AS MINIMUM/MEDIAN/MAXIMUM. TIED OPTIMA ARE RETAINED.

Dimensions	R	I	H
1	1/1/3	1/1/47	1/1.5/8
2	1/6/16	1/3/18	1/2/9
3	1/18.5/45	1/14/22	1/4.5/9
4	6/45/53	1/21/22	1/6/9
5	28/54/64	14/22/22	5/9/9
6	64/64/64	22/22/22	9/9/9

The budget example selects the lowest observed mean loss within each profile's point budget. The artifact identifies the portfolios and their burdens. Another objective or a burden constraint can favor another choice; zero observed events do not imply zero risk.

Tied minima explain the large intermediate single-objective maximum. Adding an objective can break ties and remove a previously efficient candidate, so cardinality need not increase with dimension.

Appendix C. Interpretation and Reproduction

Each registered assertion specifies a source, population, expected groups, statistic, unit, quantifier, relation, and bound. Missing groups or unsupported operations stop verification. The eight correct assertions, three historical contradictions, and eight single-field mutations were constructed after reviewing the manuscript. They test specified behavior rather than estimate general detection accuracy.

For paired mean-loss comparisons, let D_{hi} be the candidate difference in entry stratum h, with weight $w_h = n_h/n$. Then

$$\widehat{\mathrm{Var}}(\widehat{\Delta}) = \sum_h w_h^2 s_{D,h}^2/n_h, \quad s_{D,h}^2 = s_{a,h}^2 + s_{b,h}^2 - 2s_{ab,h}. \tag{8}$$

Setting $v_h = w_h^2 s_{D,h}^2/n_h$ gives approximate degrees of freedom $(\sum_h v_h)^2/\sum_h[v_h^2/(n_h-1)]$ for the family-adjusted paired t intervals.

Original raw banks and protocol digests remain unchanged. Added analyses start from committed clean source states, with code, environment, and file hashes recorded in manifests. The reproduction guide documents commands and historical provenance, including an earlier dirty source state. Automated checks cover specified relationships; an unregistered interpretation can still escape them.

Appendix D. Coefficient Ranges and Evidence Scope

The ranges in Table 12 enter the original generator as independent uniform coefficient draws per scenario. The conditional analysis selects empirical thirds within each entry category. Their realized cutoffs need not coincide with one-third and two-thirds of the declared range. Released membership tables identify every scenario, its entry category, and its selected coefficient value, allowing the subsets to be reconstructed.

The analysis applies the same sequence to every coefficient: construct paired subsets; recompute the objective matrix; enumerate the frontier; and compare membership with the full bank. It also recomputes endpoint guarantees and shared-price guarantees at $\rho = 0.50$. Per-candidate objectives and all matched-subset counts are released. This records gains as well as losses and prevents a low Jaccard from being mistaken for a uniform degradation in all portfolio outcomes.

The price proof does not extend to these coefficients. Prices enter the objective table linearly through known upgrade features. Propagation, detection, isolation, service thresholds, and restoration interact through nonlinear trajectories. A coefficient interval is therefore not an objective interval unless valid trajectory bounds are supplied. Conditional subsets provide empirical sensitivity evidence, but neither their extrema nor their intersection of frontiers certifies the continuous coefficient box.

Table 13 also suggests a practical sequence for reuse. First establish the decision set and preserve pairing in the objective table. Then separate uncertainty that changes outcome definitions, valuation, sampling, and mechanisms. Finally, match any substantive recommendation to evidence about the modeled setting. The software can expose a gap in that chain; it cannot supply missing operational measurements.

TABLE 12. ALL COEFFICIENT REGIMES. RANGES ARE ORIGINAL DECLARED ASSUMPTIONS. EACH PROFILE ENTRY GIVES LOW-THIRD / HIGH-THIRD FRONTIER JACCARD RELATIVE TO ITS FULL CONFIRMATION FRONTIER. THE REGIMES USE EQUAL ENTRY WEIGHTS AND THE SAME 130 SCENARIOS ACROSS EVERY CANDIDATE IN THAT PROFILE.

Coefficient	Declared range	Resource-constrained	Intermediate	High-capacity
Traversal rate	0.07–0.2	0.736 / 0.537	0.708 / 0.864	0.600 / 0.778
Exploit patch effect	0.5–0.95	0.650 / 0.535	0.875 / 0.750	0.667 / 0.900
Vendor patch effect	0.05–0.45	0.797 / 0.671	1.000 / 0.833	1.000 / 0.429
Identity breach multiplier	1.1–2.5	0.730 / 0.726	1.000 / 0.913	0.778 / 0.667
Identity coverage	0.6–0.95	0.686 / 0.700	0.846 / 0.739	0.700 / 1.000
Identity effect	0.4–0.9	0.634 / 0.767	0.875 / 0.909	1.000 / 0.700
Isolation success	0.4–0.85	0.815 / 0.764	0.833 / 1.000	0.545 / 0.667
Restoration throughput	0.002–0.015	0.729 / 0.676	0.957 / 1.000	0.500 / 0.818
False-positive rate	0.0002–0.002	0.735 / 0.753	0.875 / 0.818	0.889 / 0.545
Backup isolation lapse	0.02–0.25	0.754 / 0.803	1.000 / 0.864	0.545 / 0.778

TABLE 13. WHAT EACH EVIDENCE LAYER ESTABLISHES. THE LAYERS ADDRESS DIFFERENT QUESTIONS; SUCCESS IN ONE CANNOT SUBSTITUTE FOR EVIDENCE IN ANOTHER.

Evidence layer	Question answered	Boundary of the conclusion
Finite objective boxes	Can any allowed objective realization remove a candidate?	Exact for the declared Cartesian box and finite candidate set.
Shared prices	Can a common admissible tariff schedule enable a dominator?	Preserves common component prices; analyst-chosen ranges.
Paired bootstrap	How often do original certificates recur after resampling?	Conditional frequencies; no claimed frontier-confidence coverage.
Mean bounds	Can every rival be excluded using screened population coordinates?	Sufficient screen; finite-sample or approximate coverage as labeled.
Coefficient regimes	Does the complete frontier change across observed input regimes?	Conditional subsets; no fixed-coefficient intervention or joint-box guarantee.
Restoration alternative	Does a different recovery mechanism change outcomes and membership?	Paired structural comparison within a selected candidate bank.
CIPHER coverage	Which reported services and time periods are missing?	Public coded harms with repeated sources; no control-effect calibration.
Engineering benchmarks	Do finite-box calculations work on independently defined functions?	Implementation checks; no hospital validation.
Interpretation assertions	Do specified statements follow from registered data and scope?	Known-case regression checks; no general prose-verification claim.